

Program : Diploma in Chemical Engineering	
Course Code : 6071B	Course Title : Solid Waste Management
Semester : 6	Credits : 4
Course Category : Program Elective	
Periods per week : 4 (L: 4 T: 0 P: 0)	Periods per semester : 60

Course Objectives:

- To enable the students to interpret the environmental and health issues by in-adequate solid waste management.
- To identify the methods for solid waste management.

Course Prerequisites:

Topic	Course code	Course name	Semester
Soil pollution, solid waste management		Environmental Science	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize solid waste, its classification, characteristics and sources.	12	Understanding
CO2	Summarize the storage and transportation of solid waste.	18	Understanding
CO3	Summarize the methods for disposal of solid waste.	18	Understanding
CO4	Summarize the regulations for solid waste management.	12	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					3		
CO2					3		
CO3					3		
CO4					3		

3-Strongly Mapped 2-Moderately Mapped 1- Weakly Mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Summarize solid waste, its classification, characteristics and sources.		
M1.01	Outline solid waste management	2	Understanding
M1.02	Summarise the classification of solid waste	3	Understanding
M1.03	Outline Municipal solid waste	3	Understanding
M1.04	Summarise the characterization of municipal solid waste	4	Understanding
Contents: Solid waste- definition - need for solid waste management, - flow of materials and waste in an industrial society. Categories of solid waste - Quantities and Composition - Physical, Chemical and Biological Characteristics. Municipal solid waste- definition – sources Characterization of municipal solid waste - Common Components - Chemical Properties- Ultimate Analysis, Proximate Analysis, Energy Content, Fusion Point, Biodegradability, Physical Properties- Density, Moisture Content, Particle Size Distribution, Field Capacity, Impacts due to improper solid waste management.			
CO2	Summarize the storage and transportation of solid waste.		
M2.01	Identify solid waste collection systems.	3	Understanding
M2.02	Summarise transfer and transport operations	4	Understanding
M2.03	Summarise material recovery and recycle operations.	4	Understanding

M2.04	Outline the segregation methods.	6	Understanding
	Series Test – I	1	
Contents: Solid waste collection systems - Primary collection and secondary collection - Hauled container system - Stationary container system - Collection from residents, apartments, and industries. Storage - Onsite and offsite storage Transfer and Transport - Transfer stations - Open tipping floor - Open pit design - Direct transfer stations - Transfer station with compartment Materials Recovery and Recycling - components of solid waste management- Reduce, recycle, reuse and recover Segregation at different stages - segregation methods - Hand Sorting, Screens, Trommel, Air Classifiers, Float Separators, Optical Sorting, Sorting by Differential Melting Temperature, Sorting by Selective Dissolution, Magnetic Separation, Eddy Current Separators, Electrostatic Separators Shredding, Pulping, Crushing, Baling			
CO3	Summarize the methods for disposal of solid waste.		
M3.01	Differentiate aerobic and anaerobic methods for the disposal of solid waste.	3	Understanding
M3.02	Summarise the composting methods for solid waste disposal.	3	Understanding
M3.03	Summarise land fill method of solid waste disposal	4	Understanding
M3.04	Summarise common reuse and recycling opportunities for solid waste.	4	Understanding
M3.05	Summarise thermal methods of solid waste disposal	4	Understanding
Contents: Waste treatment methods- Aerobic digestion of solid waste - Anaerobic digestion of solid waste. Composting - preparation, decomposition, and product preparation - Windrow-based composting and in vessel composting, factors affecting composting of waste - Vermi composting - Factors Affecting the Composting Process- Preprocessing of the Feedstock, Environmental Factors, Nutrients, C:N Ratio, Aeration, Moisture Content, Temperature, pH Disposal - Landfill types - open dumps, controlled dumps or sanitary landfills/ engineered landfill – Stages of landfill - planning, site selection, site preparation, landfill bed construction, leachate and gas collection system incorporation, land filling, monitoring, closure of landfill, and post closure monitoring. Landfill methods - trench method, area method and depression/ Canyon method - Processes Within a Landfill - Controlling Leachate and Gas, landfill sealants, Landfill capping - Factors to be considered in landfill design -			

Factors to be considered during construction and operation of landfill - methods of venting landfill gases: (a) cell, (b) barrier, (c) well. Road Making, Plastic Granulating, Metal Removal and Recovery Incineration - Waste Heat Recovery – Gasification - Plasma Technology – Pyrolysis			
CO4	Summarize the regulations for solid waste management.		
M4.01	Identify the environmental impacts of solid waste	3	Understanding
M4.02	Summarise Pollution control methods during solid waste disposal.	3	Understanding
M4.03	Summarise hazardous waste management.	3	Understanding
M4.04	Outline laws regarding disposal of solid waste.	2	Understanding
	Series Test – II	1	
Contents: Health and Safety Issues - Safety During Transportation- Safety and environmental Issues Air Pollution Control in incinerator- The Electrostatic Precipitator, Fabric Filters, Gas Washing Zero Waste: Concept and Practice Hazardous waste – definition and classifications - Biomedical waste- classes, significance - pathogenic virus and associated disease, pathogenic bacteria and associated disease - Categories, Collection, color coding, waste segregation, Treatment and Disposal - Biomedical waste incinerator, Autoclaving, Microwaving, Deep Burial, Commonly used chemical disinfectants- Chlorine dioxide, Sodium hypochlorite, Formaldehyde, Ethylene oxide - Radioactive Waste, Treatment and conditioning, Precautions for handling radioactive waste Hazardous Wastes (Management and Handling) Rules, 1989 - Biomedical Waste (Management and Handling) Rules (1998) - Hazardous Wastes (Management, Handling and Transboundary movement) Rules, 2008 - e-waste (Management and Handling) Rules 2011			

Text / Reference

T/R	Book Title/Author
T ₁	Solid Waste Management_ Principles and Practice ; Ramesha chandrappa, Diganta Bhusan Das
R ₁	Hand book of Solid-Waste-Management ; George Tchobanoglous, Frank Kreith -
R ₂	Waste management practices; John Pichtel
R ₃	Solid Waste Management Engineering ; John T Pfeffer
R ₄	Waste Treatment and Disposal; Paul T Williams

Online Resources

S.No	Website Link
1	https://nptel.ac.in/courses/120/108/120108005/